Generalization of CAM in General Model

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Abstract

Class Activation Mapping (CAM) is a widely used technique for visualizing class-discriminative regions in convolutional neural networks, but it requires architectural modification of the target model by inserting a global average pooling (GAP) layer. Gradient-weighted CAM (Grad-CAM) has been proposed as a more general, gradient-based alternative that can be applied to a broader family of architectures without retraining. This paper provides a formal analysis of the relationship between CAM and Grad-CAM and shows that, for GAP-based CNNs, Grad-CAM reduces to CAM. Starting from the standard GAP formulation of CAM, it is proved that the Grad-CAM channel weights coincide with the CAM classifier weights up to a constant normalization factor when both methods are applied to the last convolutional layer, thereby establishing Grad-CAM as a generalized formulation of CAM. The theoretical result is complemented by an empirical study on the Stanford Dogs dataset using a ResNet-50 classifier, which already includes a GAP layer. For this model, CAM and Grad-CAM produce activation maps that are visually indistinguishable; quantitatively, the mean absolute difference between the two maps is on the order of 10^{-7}*, and the intersection over union between their derived bounding boxes is consistently 1.0. These findings clarify that, for GAP-based architectures, Grad-CAM does not yield systematically different explanations from CAM, but instead recovers CAM as a special case while retaining the flexibility to handle more general network designs.*

1. Introduction

CAM(Class Activation Map) is one of the earliest and widely used techniques in explainable AI (XAI) for convolutional neural networks (CNN), as it reveals which image regions a model relies on to decide a prediction [7]. The resulting heatmap highlights the spatial locations that contribute the most strongly to the prediction and provides an intuitive visual explanation of the decision of the model [7].

However, in order to apply CAM, the model architecture must be modified to include a GAP layer [7]. Architecture modification enforcement has been given rise to considerable controversies over the past few years. First, The architecture modification enforcement implies that there is no formal guarantee that the modified CAM network remains identical, in terms of its learned decision function, to the original target model intended for analysis [1]. Consequently, in the worst case, CAM can highlight a region of interest (RoI) that differs from the RoI of the original architecture [4]. In addition, replacing fully connected layers by GAP introduces extra training cost whenever a existing model must be adapted for the purpose of explanation. Second, the use of a GAP layer can entail a loss of spatial information. By construction, GAP aggregates each feature map into a scalar, thereby discarding the fine-grained spatial structure originally present in the convolutional activations [1, 4]. Several empirical studies have reported performance degradation or overly coarse localization when CAM is applied in such settings [4]. Finally, classical CAM is inherently restricted to the last convolutional layer. thus, explanation maps cannot be computed for earlier layers or for intermediate representations, which further limits its flexibility as a general analysis tool [4].

To address these limitations, a gradient-based generalization of CAM, termed Grad-CAM, was proposed as a model-agnostic visualization method for general deep neural networks. Unlike CAM, Grad-CAM use gradients with respect from the output [4]. Moreover, Grad-CAM operates on pre-trained networks without any architectural modification or retraining, and it is applicable to a broad family of models [4]. In this paper, a formalization of Grad-CAM as a generalized formulation of CAM is presented. Building on this formal connection, an extensive experimental investigation is conducted to examine whether Grad-CAM and CAM exhibit meaningfully different localization behavior, with respect to both qualitative visualization characteristics and quantitative localization metrics on standard image classification metrics.

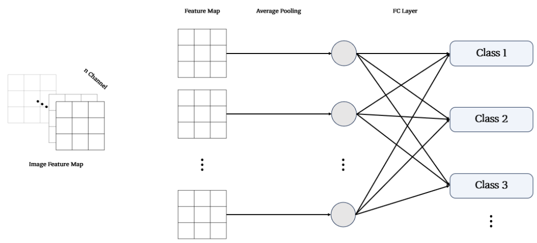

Figure 1. Illustration of a conventional CNN classifier with global average pooled feature maps and a FC layer producing class scores. Each channel of the last layer feature map is pooled into a scaler, which is then linearly combined by the FC layer to obtain the scores for different classes.

2. Literature Review

CAM is a simple and influential technique that turns an image classifier into a weakly supervised object localizer by modifying its architecture [7]. The fully connected (FC) layers at the top of a standard CNN are removed and replaced with a GAP layer followed by a linear classifier operating on the last convolutional feature maps [7]. For a given class c, the class activation map is obtained by taking a weighted sum of these feature maps where the weights are precisely the learned classifier parameters for class c [7]. This produces a class-specific spatial activation map that highlights the image regions most responsible for the prediction, which can then be interpolated and overlaid on the input image to provide a visual explanation [7]. For weakly supervised object localization with bounding box, CAM can be converted into bounding-box predictions by applying a simple post-processing scheme to the class activation maps [7]. Given the CAM for a predicted class, the heatmap is first normalized and thresholded at a fixed proportion of its maximum value to obtain a binary segmentation mask. The regions whose activation exceeds this threshold form one or more connected components, and a tight axis-aligned bounding box is drawn around the largest connected component [3, 7]. Hence, CAM-based GAP models enable to achieve competitive classification accuracy while providing strong weakly supervised localization performance despite being trained only with image-level labels and no bounding-box annotations [7]. Overall, CAM demonstrates that a standard image classification CNN, augmented only with a GAP layer, can simultaneously perform recognition and discriminative localization without any explicit localization supervision.

Grad-CAM was introduced to overcome these limitations by employing gradients as data-dependent importance weights over convolutional feature maps without imposing any architectural modification [4]. By using the gradients of a chosen output with respect to intermediate convolutional activations, Grad-CAM yields class-discriminative localization maps for off-the-shelf CNN-based models that including standard classification networks [4]. The Grad-CAM formulation strictly generalizes CAM. for fully convolutional GAP-based architectures, CAM can be recovered as a special case corresponding to a particular choice of gradient-derived weights [4]. In addition, by decoupling localization from architectural constraints, Grad-CAM can be compared against other weakly supervised localization and attention mechanisms ,for example, occlusion-based perturbation methods and probabilistic formulations while typically offering better efficiency and competitive or superior localization quality on standard benchmarks [4]. For weakly supervised object localization with boundary box, the resulting Grad-CAM heatmaps are turned into bounding-box predictions by applying the same simple post-processing as CAM so that localization quality can be directly evaluated under the standard ILSVRC protocol [3, 4]. This line of work situates our formal analysis and empirical comparison of CAM and Grad-CAM within a broader effort to design class-discriminative, architecture-agnostic visual explanations for deep CNNs.

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3. Method

3.1. Equation of CAM and Grad-CAM

This subsection recalls the standard mathematical formulation of Class Activation Mapping (CAM) and its relation to the underlying CNN architecture [7]. Let $f_k(i,j)$ assume the activation of the k-th feature map at spatial location (i,j) in the last convolutional layer of a CNN, where $k \in \{1,\ldots,K\}$ indexes channels and (i,j) indexes spatial positions. A GAP layer aggregates each feature map into a scalar

$$F_k = \frac{1}{Z}\sum_{i,j} f_k(i,j), \tag{1}$$

where Z is the number of spatial locations in the feature map.

On top of the GAP layer, a linear classifier is applied to obtain the score for class c as

$$L^C = \sum_{k=1} w_k^c F_k, \tag{2}$$

where w_k^c denotes the classifier weight connecting the k-th pooled feature F_k to class c. Substituting the definition of F_k into the class score yields

$$L^C = \sum_{k=1} w_k^c \left(\frac{1}{Z}\sum_{i,j} f_k(i,j)\right) = \frac{1}{Z}\sum_{i,j}\left(\sum_{k=1}^{K} w_k^c f_k(i,j)\right). \tag{3}$$

The term inside the parentheses can be interpreted as a class-specific spatial importance at location (i, j). Accordingly, the class activation map for class c is defined as

$$L^C = \frac{1}{Z}\sum_{i,j}\left(\sum_{k=1}^{K} w_k^c f_k(i,j)\right). \tag{4}$$

Grad-CAM Grad-CAM relaxes the architectural constraint underlying CAM by replacing the fixed classifier weights w_k^c with data-dependent importance weights computed from gradients [4]. Given a target class score S_c and the feature maps $\{A^k\}_{k=1}^{K}$, Grad-CAM first aggregates the gradients of S_c with respect to A^k over spatial locations,

$$\alpha_c^k = \frac{1}{Z}\sum_{i,j}\frac{\partial y^c}{\partial f^k(i,j)}, \tag{5}$$

and then forms the Grad-CAM map as

$$L_c = \mathrm{ReLU}\left(\sum_{k=1}\alpha_c^k f_{(i,j)}^k\right), \tag{6}$$

where ReLU discards negative contributions. Comparing (4) and (6) shows that Grad-CAM recovers CAM as a special case when the importance weights α_c^k coincide with the classifier weights w_k^c induced by a GAP-based architecture, while remaining applicable to arbitrary pretrained CNNs that contain convolutional layers.

3.2. Relationship between CAM and Grad-CAM

This subsection elucidates the theoretical relationship between CAM and Grad-CAM on the basis of the formulations presented in Section 3.1. As evidenced by (4) and (6), both methods construct a class-specific importance map by forming a weighted linear combination of convolutional feature maps. The essential distinction lies in the mechanism by which the channel-wise importance weights are determined. CAM employs the fixed classifier parameters w_k^c induced by a GAP-based linear head, whereas Grad-CAM replaces these architecture-dependent coefficients with input- and output-dependent weights α_c^k computed from gradients [4, 7]. Formally, in a GAP-based architecture the class score for class c can be written as

$$y^c = \sum_{k_n} w_{k_n}^c \frac{1}{Z}\sum_{i_n}\sum_{j_n} f_{k_n}(i_n, j_n) \tag{7}$$

where $f_{k_n}(i_n, j_n)$ denotes the activation of the k-th feature map at spatial location (i, j) in n-th layer and Z is the number of spatial locations and y^c refer to L^c at (4). The equation (7) can be computed the partial derivative of y^c with respect to an individual activation $f_k(i, j)$. By linearity of differentiation,

$$\frac{\partial y^c}{\partial f_k(i,j)} = \frac{\partial}{\partial f_k(i,j)}\left(\sum_{k_n} w_{k_n}^c \frac{1}{Z}\sum_{i_n}\sum_{j_n} f_{k_n}(i_n, j_n)\right) \tag{8}$$

From equation (8), k_n, i_n and j_n in $f_{k_n}(i_n, j_n)$ can be considered identical with as k, i and j because of CAM mechanism which use last GAP layer. Differentiation with respect to an activation in an intermediate layer would break the equalities $k_n = k$, $i_n = i$, and $j_n = j$, so that the resulting derivative must be computed via the chain rule through all subsequent layers, yielding a more complicated expression. Under the assumption that the previous expression is taken at the last layer, the equation (8) is conveniently expressed by

$$\frac{\partial y^c}{\partial f_k(i,j)} = \sum_{k}\frac{1}{Z}\frac{\partial}{\partial f_k(i,j)}\left(w_k^c\sum_{i}\sum_{j} f_k(i,j)\right) \tag{9}$$

In this case, the partial derivative is equal to 1 only when the indices of $f_k(i, j)$ coincide with (i, j) and 0 otherwise, so it can be written as follows.

$$\frac{\partial y^c}{\partial f_k(i,j)} = \frac{1}{Z} w_k^c \tag{10}$$

When both sides are summed over all spatial locations (i, j), the following expression is obtained.

$$\sum_{i}\sum_{j}\frac{\partial y^c}{\partial f_k(i,j)} = \sum_{i}\sum_{j}\left(\frac{1}{Z} w_k^c\right) \tag{11}$$

w_k^c is independent from i and j, and the summation over i and j corresponds to a sum over all spatial locations in the feature map and therefore equals Z. Consequently, the expression can be written in the following form

$$\sum_{i}\sum_{j}\frac{\partial y^c}{\partial f_k(i,j)} = w_k^c \tag{12}$$

In equation (12), the term on the left hand side can be replaced by α_c^k according to the definition in equation (5). Therefore, the expression can be simplified as follows.

$$Z\alpha_c^k = w_k^c \tag{13}$$

Hence, it can be written as follows.

$$\alpha_c^k = \frac{1}{Z} w_k^c \tag{14}$$

In equation (14), Since the factor $\frac{1}{Z}$ serves merely as a normalization term. The factor can be disregarded, activation maps being subsequently normalized to lie between 0 and 1. Hence, the Grad-CAM can be regarded as a generalized formulation of CAM.

3.3. Experimental Proof with Grad-CAM and CAM

Although the section 3.2 establishes, in a mathematical sense that Grad-CAM can be regarded as a generalized formulation of CAM, an empirical validation of this relationship remains necessary. In this paper, experiments can be conducted on the Stanford Dogs dataset [2], which consists of 120 dog breeds with bounding-box annotations and corresponding class labels for image classification. A ResNet-50 model, which is well-known image classification model, is trained on the Stanford Dogs dataset [2], and the activation maps produced by CAM and Grad-CAM are subsequently compared for the trained network. Given that a GAP layer is already involved into the ResNet-50 architecture, under the assumption that Grad-CAM constitutes a generalized extension of CAM, the presence of a GAP layer in the ResNet-50 architecture implies that the activation maps produced by Grad-CAM are not expected to exhibit substantial systematic deviations from those obtained by CAM.

The experimental procedure is organized as follows.

1. Model training. training a ResNet-50 model on the Stanford Dogs dataset. Since the primary objective of this paper is to investigate discrepancies between CAM and Grad-CAM rather than to optimize classification performance. Hence, the number of training epochs is 100. To improve training efficiency and increase data diversity, data augmentation techniques such as CutMix [5] and Mixup [6] are employed during training.
2. Comparison of activation maps. For each image in the test set, class activation maps are generated using both CAM and Grad-CAM. The element-wise difference between the two activation maps is then computed. Then, the mean of difference is used as a quantitative measure of discrepancy between the two methods.
3. Comparison of bounding boxes. Since both CAM and Grad-CAM can be used to derive class-specific bounding boxes from their respective activation maps, the intersection over union (IoU) between the bounding boxes obtained from CAM and those obtained from Grad-CAM is calculated. This IoU serves as an additional metric for assessing the discrepancy between the spatial localization behavior of the two methods.

3.4. IoU

A commonly used metric for quantifying the overlap between two regions in object detection and localization is the IoU. Given a predicted region A and a reference region B, IoU is defined as

$$IoU(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{15}$$

A prediction is typically considered correct if its IoU

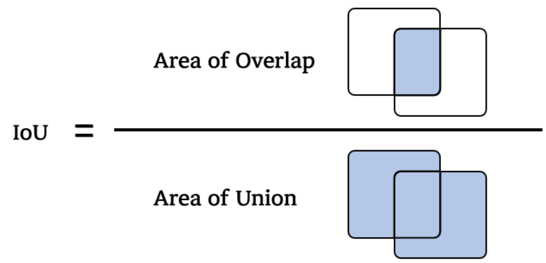


Figure 2. IoU as the ratio between the area of intersection and the area of union of two bounding boxes, illustrating how spatial agreement between predicted and reference regions is quantified.

with the ground-truth bounding box exceeds a fixed threshold, and this criterion has become standard in datasets

More generally, this quantity corresponds to the Jaccard index, which measures the similarity between two sets and is defined in the same way as the fraction of the intersection over the union. In this view, IoU can be interpreted as a specialization of the Jaccard index to the case where the sets A and B represent regions in an image.

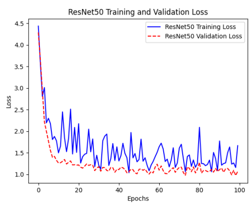
4. Result

Before the experimental results obtained on the Stanford Dogs dataset in order to empirically examine the relationship between CAM and Grad-CAM, the ResNet-50, which trained with Stanford Dogs dataset, have to be validated. As shown in Fig. 3, the model exhibits stable convergence, and Table. 1 indicates that a validation accuracy of 74.41% is achieved. Although additional training epochs could further improve the predictive performance, such optimization lies beyond the primary objective of this study. Hence, the adequately converged model at this stage is adopted for all subsequent analyses.

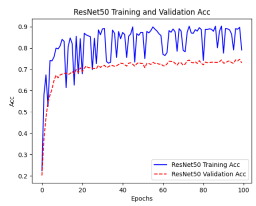
For the original input image, the activation maps from the trained ResNet-50 model obtained using CAM and Grad-CAM are shown in Fig. 4. CAM and Grad-CAM yielded no noticeable qualitative differences in the resulting activation maps. However, to investigate potential subtle discrepancies, the mean absolute difference between the two activation maps was computed for 16 samples drawn from the test set. the resulting values are summarized in Fig. 6. Although some variation is observed across images, the magnitude of these differences remains around 10^{-7}, indicating that the activation maps produced by the two methods cannot be regarded as meaningfully different. Furthermore, to assess whether the two methods exhibit consistent spatial localization, the IoU between the bounding boxes derived from CAM and Grad-CAM was computed. For all evaluated samples, an IoU value of 1.0 was obtained, indicating that the two methods produce identical bounding boxes, as illustrated in Fig. 5.

Table 1. Training and validation performance of ResNet-50 on the Stanford Dogs dataset.

	Loss	Accuracy
Training	1.0948	0.9028
Validation	1.0645	0.7441



(a) ResNet-50 trained with Stanford Dogs dataset training/validation loss

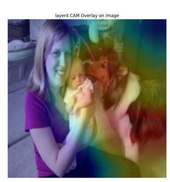


(b) ResNet-50 trained with Stanford Dogs dataset training/validation accuracy

Figure 3. Training dynamics of ResNet-50 on the Stanford Dogs dataset.(a) Loss curves over 100 epochs. (b) Accuracy curves over 100 epochs.In both plots, the blue curve denotes validation performance and the red curve denotes training performance.

(a) Original

(b) CAM



(c) Grad-CAM

Figure 4. Visualization of class-discriminative regions for a correctly classified sample from the Stanford Dogs dataset. (a) Original input image. (b) CAM-based activation map overlaid on the image. (c) Grad-CAM activation map for the same model and class, showing qualitatively similar localization behavior.

All experiments were conducted using Python 3.10.19 and PyTorch 2.9.1 with CUDA 12.8.

5. Conclusion

This work provided a formal analysis of the relationship between Class Activation Mapping (CAM) and Gradient-weighted Class Activation Mapping (Grad-CAM) within convolutional neural networks. Starting from the GAP-based formulation of CAM, the gradient-based coefficients of Grad-CAM were shown to coincide with the CAM classifier weights up to a constant normalization factor when both methods are applied to the last convolutional layer. Under this setting, Grad-CAM therefore reduces to CAM, establishing Grad-CAM as a generalized formulation that subsumes CAM for architectures equipped with a GAP layer.

The theoretical result was complemented by an empirical study on the Stanford Dogs dataset using a ResNet-50 classifier. The trained model exhibited stable convergence and reasonable validation accuracy, after which CAM- and Grad-CAM-based activation maps were compared on a testset. Visual inspection revealed no qualitative discrepancies between the two methods. Quantitatively, the mean absolute difference between the corresponding activation maps across multiple test images was on the order of 10^-7

Figure 5. Bounding boxes derived from CAM and Grad-CAM for a correctly classified sample. The two boxes coincide exactly (IoU = 1.0), resulting in a single visible bounding box and indicating identical localization between the methods.

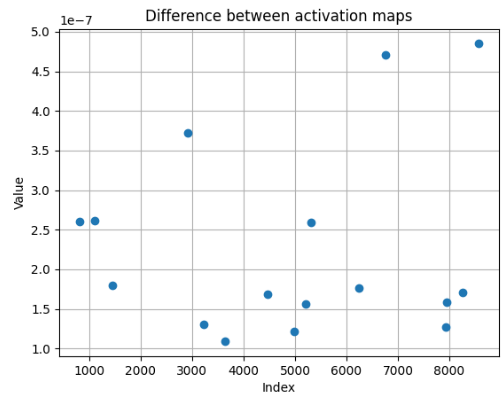


Figure 6. Mean absolute difference between CAM and Grad-CAM activation maps for 16 test samples from the Stanford Dogs dataset. Each point corresponds to a single image, with values on the order of 10^-7, indicating negligible discrepancy between the two methods.

In conclusion, these findings demonstrate that, for GAP-based CNN architectures such as ResNet-50, Grad-CAM does not provide systematically different explanations from CAM, but rather recovers CAM as a special case while retaining the flexibility to operate on more general architectures. This clarification helps delineate the regimes in which the two methods are effectively interchangeable and establishes a principled foundation for selecting gradient-based explanation techniques in practice.

6. Future Work

The limitation of the present study lies in the lack of architectural diversity. The analysis focused exclusively on ResNet-50, which already incorporates a global average pooling (GAP) layer, and thus the experiments primarily demonstrate that, for architectures equipped with a GAP layer, CAM and Grad-CAM effectively play an equivalent role. To more fully substantiate the claim that Grad-CAM constitutes a more general approach, it is necessary to investigate models that do not contain a GAP layer, derive bounding boxes using both CAM and Grad-CAM, and compare these boxes against ground-truth annotations by computing IoU for each method. Future work will therefore extend the empirical study to a broader set of network architectures and provide a more comprehensive comparative analysis in subsequent publications.

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